Training v1 Report

Overview

This document summarizes the results of the first training cycle for the BattingEdge Error Classification Model. The model was trained using 224 labeled samples across five distinct batting error categories. The primary objective of this training was to establish a baseline classifier using a Random Forest algorithm and evaluate its performance for future iterations.

Dataset Summary

A total of 224 samples were successfully loaded for the training process.

- **Feature Shape:** 224 samples × 42 features
- **Labels Shape:** 224 labels

Label Distribution

- **good:** 80 samples (35.7%)
- **head_falling:** 39 samples (17.4%)
- **late_foot:** 23 samples (10.3%)
- **low_elbow:** 27 samples (12.1%)
- **off_balance:** 55 samples (24.6%)

The dataset is moderately imbalanced, with "late_foot" and "low_elbow" having the fewest examples.

Data Splitting

The data was split into training and testing subsets.

- **Training Samples:** 179
- **Testing Samples:** 45

This split preserves class distribution while providing sufficient samples for initial evaluation.

Model Training

A Random Forest Classifier was selected for this training session.

Training Configuration

- **Number of Trees:** 200
- **Maximum Depth:** 15
- **Class Weights:** Balanced

Training completed successfully without errors.

Cross-Validation

5-fold cross-validation was performed to assess generalization.

- **Cross-Validation Accuracy:** 34.06% (±5.57%)

This indicates high variance and hints at class complexity or insufficient representative data per class.

Model Evaluation

Accuracy Scores

- **Training Accuracy:** 100.00%
- **Testing Accuracy:** 33.33%

A significant train-test gap (≈66%) was observed, indicating clear overfitting.

Classification Report

- **good:** Precision 0.42, Recall 0.62, F1 Score 0.50
- **head_falling:** Precision 0.25, Recall 0.12, F1 Score 0.17
- **late_foot:** Precision 0.00, Recall 0.00, F1 Score 0.00
- **low_elbow:** Precision 0.00, Recall 0.00, F1 Score 0.00
- **off_balance:** Precision 0.36, Recall 0.36, F1 Score 0.36

Misclassification is most prominent in minority classes. This confirms the need for more data, augmentation, or architecture adjustments.

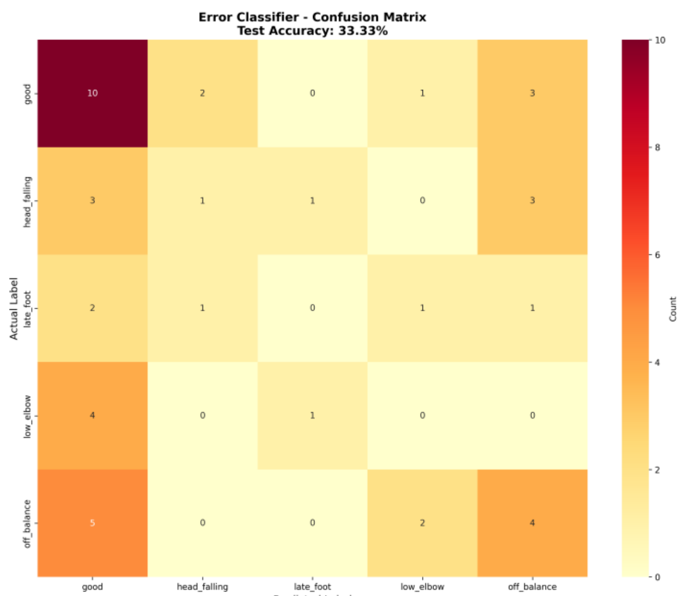
Top 10 Feature Importances

1. mean_feat5 (0.0343)

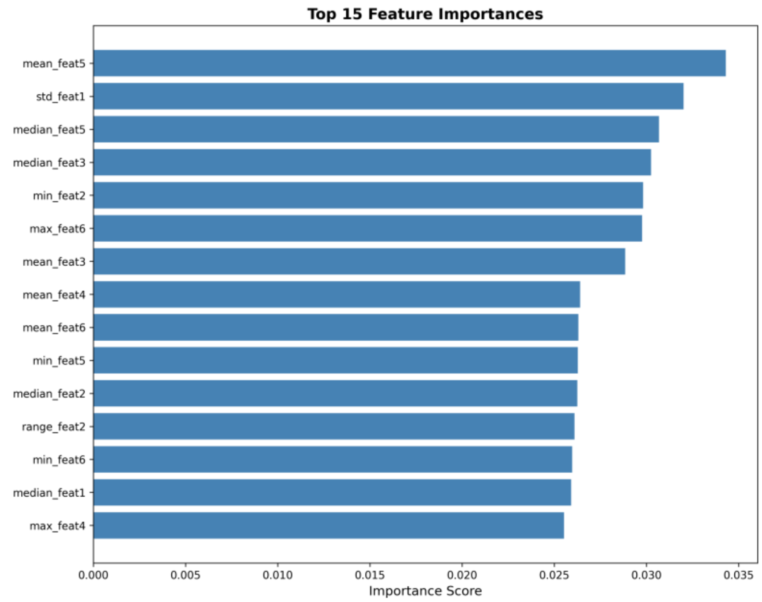
2. std_feat1 (0.0320)
3. median_feat5 (0.0307)
4. median_feat3 (0.0302)
5. min_feat2 (0.0298)
6. max_feat6 (0.0298)
7. mean_feat3 (0.0288)
8. mean_feat4 (0.0264)
9. mean_feat6 (0.0263)
10. min_feat5 (0.0263)

Generated Visualizations

- **Confusion Matrix:**



- **Feature Importance Plot:**



Summary

This marks the completion of **Training v1** for the BattingEdge Error Classification Model.

- **Final Test Accuracy:** 33.33%
- **Total Classes:** 5
- **Training Samples:** 179
- **Testing Samples:** 45

The model is functional but clearly overfitted at this stage. Future improvements should include data balancing, feature engineering, and exploration of alternative architectures.